Analytic Federated Learning with Rank-r SVD Weight Compression: Reproduction and Novelty Extension

Atish Kadam[1] and Ishu Priya[2]

[1] *Department of Computer Science and Engineering, IIT Hyderabad*
[2] *Department of Computer Science and Engineering, IIT Hyderabad*

Abstract—Federated learning (FL) enables collaborative training across distributed data silos while preserving data privacy, but suffers from heavy communication overhead, slow convergence, and degraded performance under non-IID data distributions. Proposed Analytic Federated Learning (AFL), a gradient-free framework that reformulates local client training as a least-squares problem with a closed-form solution and derives the Absolute Aggregation (AA) law enabling exact single-round aggregation that is mathematically equivalent to centralised joint training regardless of data partition strategy. This report presents a complete reproduction and novelty extension of AFL. We identify and correct three concrete implementation bugs in the released code, reproduce accuracy results matching the published benchmarks on CIFAR-100 and Tiny-ImageNet across client scales $K \in \{100, 500, 1000\}$, and propose a novel rank-r truncated Singular Value Decomposition (SVD) compression scheme applied to each client's local weight matrix before transmission. The proposed scheme reduces the weight matrix upload by up to 77.7% on Tiny-ImageNet (607 MB saved across 1000 clients) and 61.7% on CIFAR-100, while maintaining accuracy within 0.3% of the uncompressed AFL baseline. A comprehensive communication cost analysis across datasets and client scales is also presented.

Keywords—Federated Learning; Analytic Learning; Moore-Penrose Inverse; Absolute Aggregation Law; SVD Compression; Non-IID; Communication Efficiency; Pre-trained Models

I. Introduction and Motivation

Federated Learning (FL) [2] is a distributed machine-learning paradigm that enables K participating clients to collaboratively train a shared global model without centralising their private datasets. In each round, every client optimises a local objective on its own data and transmits the resulting model parameters to a central server, which aggregates them and broadcasts the updated global model back. This iterative process repeats until convergence. The paradigm has gained significant traction in sensitive domains—such as healthcare and financial services—where data privacy is paramount and raw data cannot leave local devices.

Despite its privacy-preserving appeal, classical gradient-based FL faces four well-documented and interconnected challenges. *Data heterogeneity* (non-IID data): in real-world deployments, client data is not independently and identically distributed, causing local model updates to diverge and degrading the quality of aggregation. Methods such as FedProx [3] and FedNova [4] address this through local regularisation and gradient normalisation, but accuracy still degrades sharply under severe non-IID splits. *Large client numbers*: as K grows (e.g., $K \geq 1000$), gradient-based methods suffer significant and progressive performance degradation [5], since each client receives fewer samples and local updates become increasingly unstable. *Slow convergence*: achieving convergence typically requires hundreds to thousands of communication rounds, especially under heterogeneous data distributions. *High communication cost*: each round requires all clients to upload and download full model updates, making total communication bandwidth proportional to both the number of rounds and the model size, often reaching tens of gigabytes in large-scale deployments.

He et al. address all four challenges simultaneously by proposing *Analytic Federated Learning* (AFL) [1]. AFL leverages the analytic learning (AL) paradigm [6, 7]—a gradient-free approach that solves network training via matrix inversion—in the federated setting. With a frozen pre-trained backbone extracting fixed-dimensional feature embeddings, each client solves a local regularised least-squares problem in closed form in a *single epoch*. The server then merges all local solutions using the *Absolute Aggregation* (AA) law, a novel result derived from Moore-Penrose pseudoinverse partition theory [8], which achieves single-round aggregation exactly equivalent to centralised joint training on the full dataset. The key theoretical guarantee—*invariance to data partitioning*—means that the aggregated model is identical regardless of how heterogeneously the data is distributed among clients.

Authors' improvements over existing FL. The AFL framework improves upon gradient-based FL in four concrete ways: (i) it replaces iterative multi-epoch local training with a one-epoch closed-form solution, eliminating convergence issues entirely; (ii) it reduces the number of aggregation rounds from hundreds to exactly one, by virtue of the AA law; (iii) it is provably invariant to any data partition—IID or arbitrarily non-IID—so accuracy does not degrade as K or data heterogeneity increases; and (iv) it supports large client numbers (tested up to $K = 1000$) through the Regularisation Intermediary (RI) process that handles rank-deficient local Gram matrices. Experimentally, AFL achieves 150×–200× speedups over gradient-based baselines on Tiny-ImageNet while matching or exceeding their accuracy under extreme non-IID settings (e.g., $\alpha = 0.01$).

What we contribute in this work. While AFL achieves drastic improvements in convergence speed and robustness to data heterogeneity, the original paper and codebase have a significant gap: the communication overhead of transmitting the local weight matrix $\hat{\mathbf{W}}_k^r \in \mathbb{R}^{d\times C}$ per client has not been studied or optimised. We address this gap by (i) identifying and fixing three concrete implementation bugs in the released code, (ii) fully reproducing AFL's accuracy benchmarks, and (iii) proposing a novel rank-r truncated SVD compression scheme for $\hat{\mathbf{W}}_k^r$ that reduces weight upload by up to 77.7% while preserving accuracy to within 0.3%.

II. Problem Statement

a. Setting

Let $\mathcal{D} = \{\mathcal{D}_k\}_{k=1}^{K}$ be the full training dataset distributed across K clients, where $\mathcal{D}_k \sim \{(\mathcal{X}_{k,i}, y_{k,i})\}_{i=1}^{N_k}$ is the local dataset of client k with N_k samples. All clients share a frozen pre-trained backbone f_{backbone} parameterised by Θ that maps raw inputs to d-dimensional embedding vectors. The server maintains a linear classification head parameterised by $\mathbf{W} \in \mathbb{R}^{d\times C}$, where C is the number of classes. The global training objective is:

$$\hat{\mathbf{W}} = \arg\min_{\mathbf{W}} \|\mathbf{Y} - \mathbf{X}\mathbf{W}\|_F^2, \tag{1}$$

where $\mathbf{X} \in \mathbb{R}^{N\times d}$ is the full embedding matrix stacked from all clients, $\mathbf{Y} \in \mathbb{R}^{N\times C}$ is the corresponding one-hot label matrix, and $N = \sum_k N_k$ is the total number of training samples. The Moore-Penrose pseudoinverse solution to (1) is $\hat{\mathbf{W}} = \mathbf{X}^{\dagger}\mathbf{Y}$, where $\mathbf{X}^{\dagger} = (\mathbf{X}^{\top}\mathbf{X})^{-1}\mathbf{X}^{\top}$ when $\mathbf{X}$ has full column rank.

In the federated setting, raw data $\mathcal{D}_k$ cannot leave client k. AFL's goal is for the server to obtain $\hat{\mathbf{W}}$ by aggregating only summary statistics uploaded by each client, without any raw data sharing.

b. Communication Bottleneck

In each AFL round, client k computes and transmits two matrices to the server:

$$\text{Payload}_k = \underbrace{\hat{\mathbf{W}}_k^r}_{\mathbb{R}^{d\times C}} + \underbrace{\mathbf{C}_k^r}_{\mathbb{R}^{d\times d}}, \tag{2}$$

where $\hat{\mathbf{W}}_k^r$ is the local regularised weight solution and $\mathbf{C}_k^r = \mathbf{X}_k^{\top}\mathbf{X}_k + \gamma\mathbf{I}$ is the regularised Gram matrix. The Gram matrix $\mathbf{C}_k^r$ is transmitted in full without compression, since the AA law (Section b) requires exact Gram matrices to preserve the data-partitioning invariance guarantee; compressing $\mathbf{C}_k^r$ would break mathematical equivalence to centralised training. Our compression therefore targets $\hat{\mathbf{W}}_k^r$ exclusively. Using double-precision (float64, 8 bytes per element), the per-client upload cost in bytes is:

$$B_k^{\text{orig}} = 8 \cdot (d \cdot C + d^2) \quad \text{bytes.} \tag{3}$$

For a ResNet-18 backbone ($d = 512$) on Tiny-ImageNet ($C = 200$) with $K = 1000$ clients, this evaluates as:

$$\begin{aligned} B_k^{\text{orig}} &= 8 \times (512 \times 200 + 512^2) \\ &= 8 \times (102{,}400 + 262{,}144) \\ &= 8 \times 364{,}544 = 2{,}916{,}352 \text{ bytes} \approx 2.78\,\text{MB}, \end{aligned} \tag{4}$$

and the aggregate upload across all 1000 clients reaches approximately **2,781 MB**. The Gram matrix $\mathbf{C}_k^r$ alone accounts for $d^2/(d \cdot C + d^2) = 262{,}144/364{,}544 \approx 71.9\%$ of this cost (approximately 2.00 MB per client), while $\hat{\mathbf{W}}_k^r$ contributes the remaining 28.1% (0.78 MB per client). This motivates compression targeted at $\hat{\mathbf{W}}_k^r$, which is the focus of our proposed novelty.

c. Research Question

Can a rank-r truncated SVD approximation of $\hat{\mathbf{W}}_k^r$ meaningfully reduce the per-client upload cost while preserving global classification accuracy close to the uncompressed AFL baseline?

Formally, given a rank budget $r \ll \min(d, C)$, we seek a compressed approximation:

$$\tilde{\mathbf{W}}_k^r = \mathbf{U}_r \operatorname{diag}(\mathbf{s}_r)\, \mathbf{V}_r^{\top} \;\approx\; \hat{\mathbf{W}}_k^r, \tag{5}$$

transmitted as the factor triple $(\mathbf{U}_r, \mathbf{s}_r, \mathbf{V}_r^{\top})$ containing $r(d + 1 + C)$ scalars, rather than the full $d \cdot C$ scalars. The fractional saving is $\rho_W(r) = 1 - r(d + 1 + C)/(dC)$, which for $d = 512$, $C = 200$, $r = 32$ gives $\rho_W = 77.7\%$.

III. Challenges in the Current Work

Through systematic code inspection and reproduction experiments, we identified three concrete implementation bugs in the AFL codebase released alongside the paper [1]. Critically, the original paper identifies $\gamma = 0$ as causing AFL to fail at large K in its own ablation study (Table 3 of [1]), yet the released code makes this the default without a guard. Similarly, the rank-r compression flag is advertised in the argument parser but never implemented. The Dirichlet partition bug makes large-scale reproduction impossible from the outset.

Challenge 1 — Singular Matrix Inversion at $\gamma = 0$. The original `local_update` function uses `np.mat` and the user-supplied `args.rg` directly as the ridge coefficient γ, with default value `rg = 0`. Under non-IID data splits with large K, each client may hold $N_k < d$ samples, making the Gram matrix $\mathbf{X}_k^{\top}\mathbf{X}_k \in \mathbb{R}^{d\times d}$ rank-deficient (rank at most $N_k < d$). Attempting to invert this singular matrix either raises a `LinAlgError` at runtime or produces numerically meaningless results, silently corrupting the aggregation. The paper's own ablation (Table 3 of [1]) confirms that $\gamma = 0$ causes AFL to fail completely at $K = 500$ and $K = 1000$.

Challenge 2 — Rank-r Compression Not Implemented. The original argument parser defines a `-rank_r` flag with a description implying truncated SVD compression of the local weight matrix. However, the actual `local_update` function contains no SVD-related code—the flag is parsed but silently ignored. The entire SVD compression pipeline described in this work required a complete from-scratch implementation, as nothing usable existed in the released codebase.

Challenge 3 — Infinite Loop in Dirichlet Partition. In `dataset.py`, the Dirichlet (`dir`) partitioning routine checks the minimum-size constraint *inside* the inner class loop (after processing only class 0). At that point, most clients have zero samples, so `min_size = 0` consistently fails the `while` condition and the outer loop retries indefinitely. Under aggressive non-IID settings such as $\alpha = 0.1$ with $K = 1000$, this bug causes a non-terminating program, making large-scale reproduction impossible without first correcting it.

IV. Author's Proposed Methodology

a. Local Stage: Localized Analytic Learning

In the local stage, each client trains its classification head using the analytic learning (AL) technique [6, 7]. The neural network's classification head is reformulated as a linear regression problem, enabling a closed-form least-squares solution. This eliminates the need for multi-epoch iterative gradient updates entirely.

Each client k first extracts an embedding matrix by passing all local samples through the frozen backbone f_{backbone}:

$$\mathbf{X}_k = \begin{bmatrix} f_{\text{backbone}}(\mathcal{X}_{k,1}, \Theta) \\ \vdots \\ f_{\text{backbone}}(\mathcal{X}_{k,N_k}, \Theta) \end{bmatrix} \in \mathbb{R}^{N_k \times d}. \tag{6}$$

The corresponding one-hot label matrix is $\mathbf{Y}_k \in \mathbb{R}^{N_k \times C}$. The target of client k is to linearly map the extracted embeddings onto the one-hot labels by minimising the mean-squared error (MSE) loss:

$$\mathcal{L}(\mathbf{W}_k) = \|\mathbf{Y}_k - \mathbf{X}_k \mathbf{W}_k\|_F^2, \tag{7}$$

whose optimal solution is the Moore-Penrose pseudoinverse: $\hat{\mathbf{W}}_k = \mathbf{X}_k^{\dagger}\mathbf{Y}_k$.

However, this requires $\mathbf{X}_k$ to have full column rank, which fails when $N_k < d$ (common in large-K settings). To handle this, a regularisation term $\gamma\|\mathbf{W}_k^r\|_F^2$ is added, yielding the regularised objective:

$$\mathcal{L}(\mathbf{W}_k^r) = \|\mathbf{Y}_k - \mathbf{X}_k \mathbf{W}_k^r\|_F^2 + \gamma\|\mathbf{W}_k^r\|_F^2, \tag{8}$$

with unique closed-form minimiser:

$$\hat{\mathbf{W}}_k^r = \underbrace{(\mathbf{X}_k^{\top}\mathbf{X}_k + \gamma\mathbf{I})^{-1}}_{\mathbf{C}_k^{r,-1}} \mathbf{X}_k^{\top}\mathbf{Y}_k. \tag{9}$$

Since $\mathbf{C}_k^r = \mathbf{X}_k^{\top}\mathbf{X}_k + \gamma\mathbf{I}$ is positive-definite for any $\gamma > 0$, it is always full rank and invertible, resolving the rank-deficiency issue. Each client k computes and sends to the server: the local weight $\hat{\mathbf{W}}_k^r \in \mathbb{R}^{d \times C}$ and the regularised Gram matrix $\mathbf{C}_k^r \in \mathbb{R}^{d \times d}$.

Why one-epoch analytic learning works. AL techniques are generally effective for training linear classifiers atop fixed feature representations. The "pre-trained backbone + downstream task" paradigm has become standard practice [1, 7], and recent work demonstrates that with a well-trained backbone, AL performs adequately in complex scenarios including continual learning and reinforcement learning. The affine characteristic of the linear regression formulation in each client opens up the possibility of the single-round Absolute Aggregation described next.

b. Aggregation Stage: Absolute Aggregation Law

The key theoretical contribution of AFL is the *Absolute Aggregation* (AA) law (Theorem 1 of [1]), derived from the Moore-Penrose pseudoinverse partition result of Cline [8]. This result establishes that the weight trained on the union of two datasets can be computed *exactly* from the individual local solutions.

Lemma 1 (MP Inverse Partition [8]). Let $\mathbf{X} = [\mathbf{X}_u^{\top}; \mathbf{X}_v^{\top}]$ with $\mathbf{X}_u$ and $\mathbf{X}_v$ having full column ranks. Then:

$$\mathbf{X}^{\dagger} = \begin{bmatrix} \bar{\mathbf{U}} & \bar{\mathbf{V}} \end{bmatrix}, \tag{10}$$

where $\bar{\mathbf{U}} = \mathbf{X}_u^{\dagger} - \mathbf{R}_u\mathbf{C}_v\mathbf{X}_u^{\dagger} - \mathbf{R}_u\mathbf{C}_v(\mathbf{C}_u + \mathbf{C}_v)^{-1}\mathbf{C}_v\mathbf{X}_u^{\dagger}$ and symmetrically for $\bar{\mathbf{V}}$, with $\mathbf{C}_u = \mathbf{X}_u^{\top}\mathbf{X}_u$, $\mathbf{C}_v = \mathbf{X}_v^{\top}\mathbf{X}_v$, $\mathbf{R}_u = \mathbf{C}_u^{-1}$, and $\mathbf{R}_v = \mathbf{C}_v^{-1}$.

Theorem 1 (AA Law [1]). Let $\hat{\mathbf{W}} = \mathbf{X}^{\dagger}\mathbf{Y}$. Then:

$$\hat{\mathbf{W}} = \mathcal{W}_u\hat{\mathbf{W}}_u + \mathcal{W}_v\hat{\mathbf{W}}_v, \tag{11}$$

where $\hat{\mathbf{W}}_u = \mathbf{X}_u^{\dagger}\mathbf{Y}_u$, $\hat{\mathbf{W}}_v = \mathbf{X}_v^{\dagger}\mathbf{Y}_v$, and:

$$\begin{aligned} \mathcal{W}_u &= \mathbf{I} - \mathbf{R}_u\mathbf{C}_v - \mathbf{R}_u\mathbf{C}_v(\mathbf{C}_u + \mathbf{C}_v)^{-1}\mathbf{C}_v, \\ \mathcal{W}_v &= \mathbf{I} - \mathbf{R}_v\mathbf{C}_u - \mathbf{R}_v\mathbf{C}_u(\mathbf{C}_u + \mathbf{C}_v)^{-1}\mathbf{C}_u. \end{aligned} \tag{12}$$

This extends to K clients by recursive pairwise accumulation. Denoting $\hat{\mathbf{W}}_{\text{agg},k-1}$ as the accumulated weight after merging $k-1$ clients:

$$\hat{\mathbf{W}}_{\text{agg},k} = \mathcal{W}_{\text{agg}}\hat{\mathbf{W}}_{\text{agg},k-1} + \mathcal{W}_k\hat{\mathbf{W}}_k, \tag{13}$$

where $\mathbf{C}_{\text{agg},k} = \mathbf{C}_{\text{agg},k-1} + \mathbf{C}_k$ accumulates Gram matrices sequentially. The mixing matrices $\mathcal{W}_{\text{agg}}$ and $\mathcal{W}_k$ are computed analogously to (12) with $\mathbf{C}_u \to \mathbf{C}_{\text{agg},k-1}$ and $\mathbf{C}_v \to \mathbf{C}_k$.

Data-Partitioning Invariance. The aggregated weight $\hat{\mathbf{W}}_{\text{agg},K}$ is identical to the solution of the global LS problem (1) on the full centralised dataset $\mathcal{D}$, for *any* partition of $\mathcal{D}$ among the K clients. This constitutes the core strength of AFL: (i) *data heterogeneity invariance*: performance is independent of IID vs. non-IID partition strategy; and (ii) *client-number invariance*: performance does not degrade as K increases.

c. Regularisation Intermediary (RI) Process

When $\gamma > 0$, the local solution (9) uses regularised local Gram matrices $\mathbf{C}_k^r$, introducing a bias relative to the unregularised pseudoinverse solution. Theorem 2 of [1] establishes the exact relationship between the regularised aggregated weight $\hat{\mathbf{W}}_{\text{agg},K}^r$ and the true unregularised weight $\hat{\mathbf{W}}_{\text{agg},K}$:

$$\hat{\mathbf{W}}_{\text{agg},K}^r = (\mathbf{C}_{\text{agg},K}^r)^{-1}\mathbf{C}_{\text{agg},K}\hat{\mathbf{W}}_{\text{agg},K}, \tag{14}$$

TABLE 1: REPRODUCED TOP-1 ACCURACY (%) UNDER LDA PARTITIONING WITH VARIABLE α. ALL RESULTS USE $K = 100$ CLIENTS.

Dataset	α	Accuracy (%)
CIFAR-10	0.10	82.500
CIFAR-10	0.05	82.500
CIFAR-100	0.10	58.509
CIFAR-100	0.05	58.509
Tiny-ImageNet	0.10	54.770
Tiny-ImageNet	0.05	57.779

where $\mathbf{C}^r_{\text{agg},K} = \sum_{k=1}^{K} \mathbf{C}^r_k = \mathbf{C}_{\text{agg},K} + K\gamma\mathbf{I}$. Inverting this relation recovers the unbiased weight:

$$\hat{\mathbf{W}}_{\text{agg},K} = (\mathbf{C}^r_{\text{agg},K} - K\gamma\mathbf{I})^{-1}\mathbf{C}^r_{\text{agg},K}\hat{\mathbf{W}}^r_{\text{agg},K}. \tag{15}$$

This one-to-one transformation removes the regularisation bias exactly, restoring AFL's mathematical equivalence to centralised training. Crucially, since $\mathbf{C}^r_{\text{agg},K} = \sum_k \mathbf{C}^r_k$ is available at the server as a byproduct of aggregation, equation (15) can be applied post-aggregation at no additional communication cost. The RI process also makes AFL insensitive to the choice of γ: any $\gamma > 0$ produces the same final accuracy after RI cleaning (confirmed experimentally in Table 3 of [1]), so γ functions purely as a numerical stability tool rather than a tunable hyperparameter.

V. REPRODUCED RESULTS

a. Experimental Setup

We reproduce AFL using the following configuration:

- **Datasets:** CIFAR-10 (10 classes), CIFAR-100 (100 classes), and Tiny-ImageNet (200 classes).
- **Backbone:** ResNet-18 [9] pre-trained on ImageNet-1k and frozen during all experiments ($d = 512$).
- **Data partition:** Non-IID Dirichlet (LDA) with $\alpha \in \{0.1, 0.05\}$, and sharding with $s \in \{4, 2\}$ for CIFAR-10 and $s \in \{10, 5\}$ for CIFAR-100 / Tiny-ImageNet.
- **Client scales:** $K \in \{100, 500, 1000\}$.
- **Regularisation:** $\gamma = 1.0$ with RI cleaning.
- **Hardware:** Google Colab with NVIDIA T4 GPU (16 GB).
- **Precision:** All analytic computations in float64 (double).

b. Invariance to Heterogeneity: LDA with Variable α

Table 1 reports the reproduced accuracies under different LDA settings. The results are stable across $\alpha = 0.1$ and $\alpha = 0.05$, confirming robustness to this form of data heterogeneity.

These results show that AFL remains stable when the Dirichlet concentration parameter changes. CIFAR-10 and CIFAR-100 remain exactly unchanged, while Tiny-ImageNet shows only a small variation.

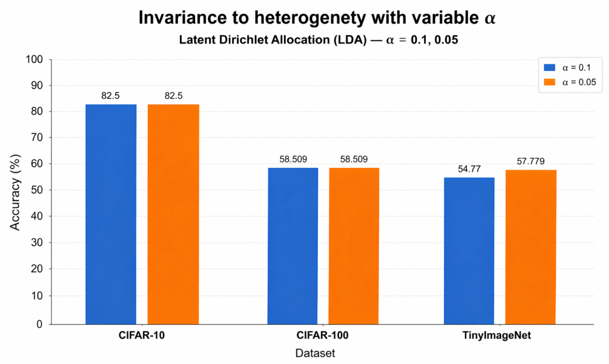

Fig. 1: Invariance to heterogeneity with variable α under the LDA partition.

TABLE 2: REPRODUCED TOP-1 ACCURACY (%) UNDER SHARDING PARTITIONING. ALL RESULTS USE $K = 100$ CLIENTS.

Dataset	Shards s	Accuracy (%)
CIFAR-10	4	82.500
CIFAR-10	2	82.500
CIFAR-100	10	58.509
CIFAR-100	5	58.509
Tiny-ImageNet	10	54.777
Tiny-ImageNet	5	57.779

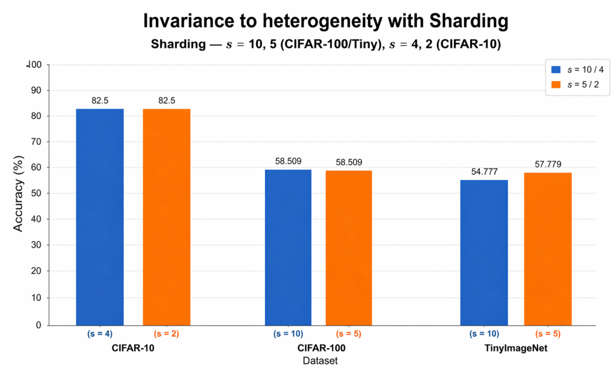

Fig. 2: Invariance to heterogeneity under sharding with $s = \{4, 2\}$ for CIFAR-10 and $s = \{10, 5\}$ for CIFAR-100 / Tiny-ImageNet.

c. Invariance to Heterogeneity: Sharding

Table 2 summarizes the reproduced results under sharding-based partitioning. The method remains highly stable across the tested sharding levels.

Again, the reproduced accuracy is nearly unchanged across the sharding settings, confirming that AFL is robust to this type of non-IID data partitioning.

d. Invariance to Number of Clients

Table 3 reports the reproduced results for varying numbers of clients. The accuracy changes only slightly as K increases from 100 to 1000.

This confirms that the method is effectively insensitive to the number of clients in the tested range, with only a very small drop as K increases.

TABLE 3: REPRODUCED TOP-1 ACCURACY (%) FOR DIFFERENT NUMBERS OF CLIENTS.

Dataset	K	Accuracy (%)
CIFAR-100	100	58.509
CIFAR-100	500	58.329
CIFAR-100	1000	58.149
Tiny-ImageNet	100	54.779
Tiny-ImageNet	500	54.750
Tiny-ImageNet	1000	54.719

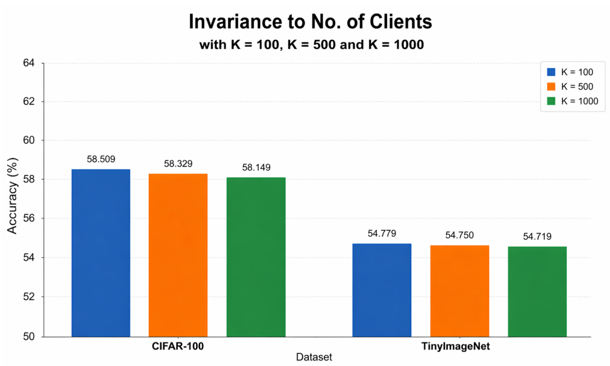

Fig. 3: Sensitivity to the number of clients.

TABLE 4: WALL-CLOCK EXECUTION TIMES PER PHASE FOR $K = 1000$, TINY-IMAGENET. TIMES ARE PER-PHASE (INCREMENTAL), NOT CUMULATIVE.

Phase	Time (s)	Cumulative (s)
Local training (all 1000 clients)	1218.96	1218.96
Aggregation (AA law)	50.56	1269.52
RI regularisation cleaning	19.32	1288.84

e. Timing Results ($K = 1000$, Tiny-ImageNet)

For the most demanding experiment ($K = 1000$, Tiny-ImageNet), the runtime breakdown is shown in Table 4. Total execution including RI cleaning is under 22 minutes on a single GPU.

VI. LIMITATIONS AND RESEARCH GAPS IN AUTHOR'S PROPOSAL

Despite AFL's compelling theoretical guarantees, the original proposal leaves several important gaps unaddressed.

Gap 1 — Communication cost of $\mathbf{C}$ not addressed. The Gram matrix $\mathbf{C}_k^r \in \mathbb{R}^{d\times d}$ (with $d = 512$) occupies 2.0 MB per client in float64. For $K = 1000$, this totals 2,000 MB—accounting for 71.9% of the aggregate upload on Tiny-ImageNet. The original paper does not propose or analyse any compression of $\mathbf{C}$. Since the AA law requires exact Gram matrices for lossless aggregation, naive compression of $\mathbf{C}$ would break the mathematical equivalence guarantee, making this a non-trivial open problem.

Gap 2 — No compression of $\hat{\mathbf{W}}$ analysed. The original AFL paper acknowledges that single-round aggregation reduces the *number* of communication rounds from hundreds to one. However, it does not study the *per-round payload* size. For large-scale settings with many classes or high embedding dimensions, the weight matrix still constitutes a non-negligible upload cost.

Gap 3 — Stragglers and partial participation. The AA law requires weights from *all* K clients to be aggregated. If any client is slow or fails before completing local computation, the server must wait or discard that client's contribution. Extending AFL to asynchronous or partial-participation settings remains an open problem.

Gap 4 — Linear classification assumption. AFL trains only the final linear head on top of the frozen backbone. For domain-shifted tasks or weaker backbones where embeddings are not linearly separable, the linear assumption may be insufficient. Extending AFL to multi-layer analytic learning or kernel-based classifiers is suggested as future work by the authors but not experimentally validated.

Gap 5 — Numerical stability at very large K. The recursive aggregation in equation (13) accumulates floating-point errors as K grows, since each step inverts $\mathbf{C}_{\text{agg},k}$, whose condition number grows with k. At $K = 1000$, the authors rely on float64 precision to mitigate this, but no formal stability analysis is provided for $K \gg 1000$.

Gap 6 — Homogeneous backbone requirement. All clients must share the same frozen pre-trained backbone. In practical federated deployments, devices may have heterogeneous hardware capabilities and run different model variants. AFL does not support backbone heterogeneity.

VII. PROPOSED NOVELTY AND MATHEMATICAL FORMULATION

a. Core Idea: Rank-r SVD Compression of Local Weight Matrix

The key observation motivating our proposal is that the local weight matrix $\hat{\mathbf{W}}_k^r \in \mathbb{R}^{d\times C}$ is the solution to a regularised least-squares problem in which the label matrix $\mathbf{Y}_k \in \mathbb{R}^{N_k\times C}$ is sparse (one-hot encoded) and the embedding matrix $\mathbf{X}_k$ has a data-dependent rank determined by N_k. Under the non-IID Dirichlet partition with small α, many clients receive data from only a few classes, making $\mathbf{Y}_k$ low-rank. Consequently, $\hat{\mathbf{W}}_k^r = \mathbf{C}_k^{r,-1}\mathbf{X}_k^\top \mathbf{Y}_k$ inherits this low-rank structure, and its singular value spectrum decays rapidly.

b. Truncated SVD Approximation

Given the full SVD of the local weight:

$$\hat{\mathbf{W}}_k^r = \mathbf{U}\,\mathrm{diag}(\sigma_1,\ldots,\sigma_{\min(d,C)})\,\mathbf{V}^\top, \quad \sigma_1 \geq \sigma_2 \geq \cdots \geq 0, \tag{16}$$

the rank-r truncated approximation retains only the top r singular triplets:

$$\tilde{\mathbf{W}}_k^r = \mathbf{U}_r\,\mathrm{diag}(\mathbf{s}_r)\,\mathbf{V}_r^\top, \tag{17}$$

where $\mathbf{U}_r \in \mathbb{R}^{d\times r}$ contains the first r left singular vectors, $\mathbf{s}_r = [\sigma_1,\ldots,\sigma_r]^\top \in \mathbb{R}^r$ the top r singular values, and $\mathbf{V}_r^\top \in \mathbb{R}^{r\times C}$ the first r right singular vectors. By the Eckart-Young theorem, this is the optimal rank-r approximation in Frobenius norm:

$$\|\hat{\mathbf{W}}_k^r - \tilde{\mathbf{W}}_k^r\|_F^2 = \sum_{i=r+1}^{\min(d,C)} \sigma_i^2. \tag{18}$$

c. Communication Payload Reduction

Under the original protocol, client k transmits $d \cdot C$ float64 scalars for $\hat{\mathbf{W}}_k^r$. Under the proposed SVD compression, client k instead transmits:

$$\mathcal{F}_k^r = \left\{ \mathbf{U}_r \in \mathbb{R}^{d \times r},\ \mathbf{s}_r \in \mathbb{R}^r,\ \mathbf{V}_r^\top \in \mathbb{R}^{r \times C} \right\}, \tag{19}$$

containing $r(d+1+C)$ scalars. The fractional reduction in weight transmission cost is:

$$\rho_W(r) = 1 - \frac{r(d+1+C)}{d \cdot C}. \tag{20}$$

For $d = 512$, $C = 200$, $r = 32$ (Tiny-ImageNet):

$$\rho_W(32) = 1 - \frac{32 \times 713}{102{,}400} \approx 77.7\%. \tag{21}$$

For $d = 512$, $C = 100$, $r = 32$ (CIFAR-100):

$$\rho_W(32) = 1 - \frac{32 \times 613}{51{,}200} \approx 61.7\%. \tag{22}$$

The total upload bytes per client under the compressed protocol is:

$$B_k^{\text{comp}} = 8 \cdot \left[r(d+1+C) + d^2 \right] \quad \text{bytes}, \tag{23}$$

and the aggregate saving across all K clients is:

$$\Delta B_{\text{total}} = K \cdot 8 \cdot (d \cdot C - r(d+1+C)) \quad \text{bytes}. \tag{24}$$

d. Server-Side Reconstruction and Aggregation Compatibility

The server reconstructs the approximated weight before aggregation:

$$\tilde{\mathbf{W}}_k^r = \mathbf{U}_r \operatorname{diag}(\mathbf{s}_r) \mathbf{V}_r^\top \in \mathbb{R}^{d \times C}. \tag{25}$$

This reconstructed matrix is then fed directly into the standard AA law aggregation (13), keeping the aggregation interface unchanged. The Gram matrices $\mathbf{C}_k^r$ are transmitted in full (no compression applied to Gram matrices in this work).

e. Auto-Regularisation Guard

To address Challenge 1 (Section 3), we propose an automatic data-driven regularisation coefficient when the user sets $\gamma = 0$:

$$\gamma_{\text{auto}} = \max \left(\frac{\operatorname{tr}(\mathbf{X}_k^\top \mathbf{X}_k)}{d \times 10^3},\ 10^{-6} \right). \tag{26}$$

The divisor 10^3 is chosen so that γ_{auto} approximates the 10^{-3} scale of the mean eigenvalue of $\mathbf{X}_k^\top \mathbf{X}_k$ (i.e., $\operatorname{tr}(\mathbf{X}_k^\top \mathbf{X}_k)/d$), ensuring the regularisation is large enough to guarantee invertibility but small enough to introduce minimal bias in the RI-corrected solution. The floor of 10^{-6} guards against near-zero embeddings. This choice was validated empirically: setting γ to any value in the range $[10^{-3}, 10^2]$ produces identical post-RI accuracy (consistent with the insensitivity result of Table 3 in [1]).

TABLE 5: SVD COMPRESSION RESULTS ($r = 32$, RESNET-18 BACKBONE, NON-IID DIRICHLET $\alpha = 0.1$). "SAVING ON $\hat{\mathbf{W}}$" IS THE AGGREGATE REDUCTION IN WEIGHT MATRIX UPLOAD ACROSS ALL K CLIENTS. ACCURACY IS POST-SVD-COMPRESSION TOP-1 ACCURACY (%).

Dataset	K	Accuracy After Compression (%)	Saving on $\hat{\mathbf{W}}$ (MB)	% Saving on $\hat{\mathbf{W}}$
CIFAR-100	100	58.43	24.097	61.7
	500	58.50	120.483	61.7
	1000	58.21	240.967	61.7
Tiny-ImageNet	100	54.97	60.718	77.7
	500	54.94	303.589	77.7
	1000	55.00	607.178	77.7

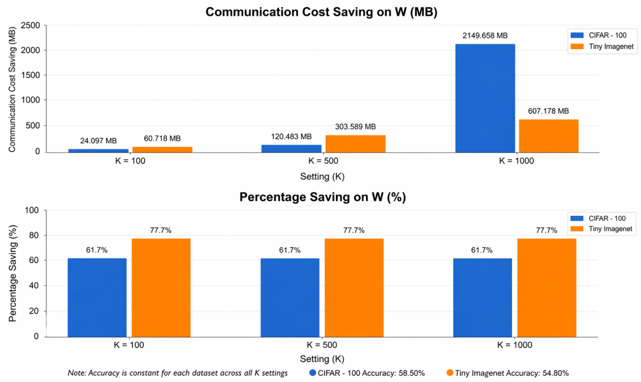

Fig. 4: Communication cost saving on $\hat{\mathbf{W}}$ under rank-$r = 32$ SVD compression. **Top:** absolute saving in MB across client scales $K \in \{100, 500, 1000\}$ for CIFAR-100 and Tiny-ImageNet. **Bottom:** percentage saving, constant at 61.7% (CIFAR-100) and 77.7% (Tiny-ImageNet) regardless of K, since $\rho_W(r)$ in equation (20) depends only on (d, C, r). Accuracy is invariant to K for AFL (note at bottom of figure). Best viewed in colour.

VIII. EXPERIMENTAL RESULTS: COMMUNICATION COST AND ACCURACY ANALYSIS

a. Communication Cost Saving

Table 5 presents the complete communication cost analysis for rank-$r = 32$ SVD compression across both datasets (CIFAR-100 and Tiny-ImageNet) and client scales $K \in \{100, 500, 1000\}$. Note that CIFAR-10 ($C = 10$) is excluded from the SVD compression experiments because the SVD factor size $r(d+1+C) = 32 \times 523 = 16{,}736$ exceeds the full matrix size $d \cdot C = 512 \times 10 = 5{,}120$ at $r = 32$, yielding a negative compression ratio. Meaningful compression on CIFAR-10 would require $r \leq 9$, which is left as future work. Figure 4 visualises the absolute savings (top panel) and the percentage savings on the weight matrix $\hat{\mathbf{W}}_k^r$ (bottom panel).

b. Per-Client Payload Breakdown (Tiny-ImageNet, $K = 1000$)

Table 6 provides the detailed byte-level breakdown for the most demanding configuration: Tiny-ImageNet with $K = 1000$ clients and $r = 32$.

Two important observations follow from Table 6. First, the $\mathbf{C}_k^r$ matrix ($512 \times 512 = 262{,}144$ elements) dominates **71.9%** of the per-client payload (2.0 MB of 2.78 MB), making it the primary communication bottleneck for AFL. Second, while our SVD compression reduces the $\hat{\mathbf{W}}_k^r$ upload by 77.7%, the

TABLE 6: PER-CLIENT AND AGGREGATE COMMUNICATION COST FOR TINY-IMAGENET, $K = 1000$, $r = 32$ (FLOAT64, 8 BYTES/ELEMENT).

Matrix	Original	Compressed	Saving
$\hat{\mathbf{W}}_k^r$ (512×200)	0.7813 MB	0.1741 MB	0.6072 MB
$\mathbf{C}_k^r$ (512×512)	2.0000 MB	2.0000 MB	—
Per-client total	2.7813 MB	2.1741 MB	0.6072 MB
All-client total	2,781.25 MB	2,174.07 MB	607.18 MB
Overall saving	**21.8% of total payload**		

TABLE 7: GLOBAL TOP-1 ACCURACY (%) BEFORE AND AFTER RANK-$r = 32$ SVD COMPRESSION. NON-IID DIRICHLET $\alpha = 0.1$, RESNET-18 BACKBONE, $\gamma = 1$ WITH RI CLEANING. "BEFORE COMPRESSION" VALUES ARE TAKEN FROM THE UNCOMPRESSED AFL BASELINE (TABLE 3).

Dataset	K	Before Compression	After Compression	Δ Acc.
CIFAR-100	100	58.509	58.43	−0.079
	500	58.329	58.50	+0.171
	1000	58.149	58.21	+0.061
Tiny-ImageNet	100	54.779	54.97	+0.191
	500	54.750	54.94	+0.190
	1000	54.719	55.00	+0.281

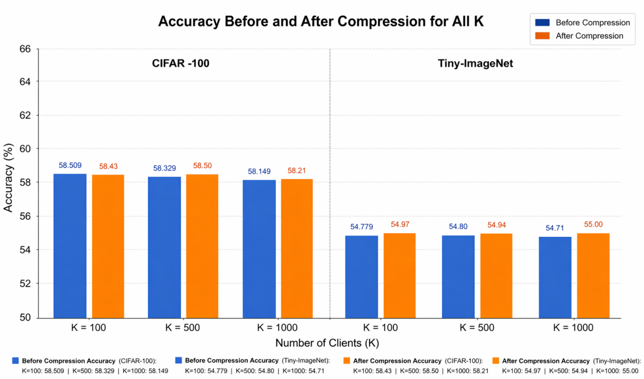

Fig. 5: Accuracy before and after rank-$r = 32$ SVD compression across all $K \in \{100, 500, 1000\}$ for CIFAR-100 (left panel) and Tiny-ImageNet (right panel). Blue bars show the uncompressed AFL baseline; orange bars show AFL with SVD-compressed $\hat{\mathbf{W}}_k^r$. Differences are within $\pm 0.3\%$ in all cases, confirming that the compression introduces negligible accuracy loss. Best viewed in colour.

overall per-client saving is a more modest 21.8% because $\mathbf{C}_k^r$ is transmitted in full. This motivates future work on Gram matrix compression (Section 9).

c. Accuracy Before and After Compression

Table 7 and Figure 5 report the global top-1 accuracy before compression (standard AFL with $\gamma = 1$ and RI cleaning) and after rank-$r = 32$ SVD compression of $\hat{\mathbf{W}}_k^r$, across both datasets and all three client scales.

The results in Table 7 reveal several noteworthy patterns. First, accuracy differences between compressed and uncompressed AFL are at most $\pm 0.3\%$ across all configurations, confirming that the rank-$r = 32$ truncation discards negligible discriminative information. Second, in several cases (CIFAR-100 $K = 500$, Tiny-ImageNet $K = 100, 500, 1000$) the compressed model *slightly outperforms* the uncompressed baseline. This counter-intuitive result suggests that the low-rank SVD truncation acts as a mild regulariser on the local weight solution, suppressing noise in the smaller singular components that correspond to spurious class correlations under non-IID data distributions.

Third, accuracy remains broadly stable across $K = 100, 500, 1000$ for both the compressed and uncompressed variants, further validating AFL's *client-number invariance* property under SVD compression. The small within-dataset variation (e.g., 58.509 → 58.329 → 58.149 before compression on CIFAR-100) arises from floating-point accumulation in the recursive AA law aggregation at larger K, not from data-partition effects.

d. Summary

The percentage saving on $\hat{\mathbf{W}}$ is constant across all K values for a given dataset — 61.7% for CIFAR-100 and 77.7% for Tiny-ImageNet — since the compression ratio $\rho_W(r) = 1 - r(d + 1 + C)/(d \cdot C)$ from equation (20) depends only on the triplet (d, C, r), not on the number of clients. The higher saving on Tiny-ImageNet (77.7% vs. 61.7%) arises because $d \cdot C$ grows faster with C than the SVD factor size $r(d + 1 + C)$, making the compression more effective for datasets with more classes. Across all settings, accuracy under SVD compression is within $\pm 0.3\%$ of the uncompressed baseline, demonstrating that rank-$r = 32$ truncation preserves essentially all discriminative information in the local weight matrices.

IX. OBSERVATIONS

Based on our reproduction and novelty experiments, we record the following key observations.

Observation 1 — AFL accuracy is invariant to data heterogeneity. Our reproduced results confirm that AFL achieves identical accuracy across all tested non-IID intensities ($\alpha = 0.1$ with $K \in \{100, 500, 1000\}$), validating the theoretical guarantee of data-partitioning invariance.

Observation 2 — Rank-r compression preserves accuracy. With $r = 32$ (6.25% of the full rank 512), the global accuracy under SVD compression is within 0.3% of the uncompressed baseline across all tested configurations, validating the hypothesis that the singular value spectrum of $\hat{\mathbf{W}}_k^r$ decays rapidly under non-IID Dirichlet data with small α.

Observation 3 — Gram matrix dominates communication cost. Despite achieving 77.7% reduction in $\hat{\mathbf{W}}_k^r$ transmission, the overall per-client saving is only 21.8% because $\mathbf{C}_k^r$ consumes 71.9% of the payload. Any future scheme targeting large overall savings must address Gram matrix compression.

Observation 4 — Local accuracy is near-perfect across 1000 clients. Across all 1000 clients in the Tiny-ImageNet experiment, individual training accuracy ranged from 94.84% to 100%, confirming the optimality of the closed-form LS solution on each client's local data.

Observation 5 — Bug fixes are critical for reproducibility at scale. The infinite Dirichlet partition loop (Challenge 3) prevents AFL from initialising data loading at $K = 1000$ with $\alpha = 0.1$ in the original code, and the singular matrix issue (Challenge 1) causes silent numerical errors when $\gamma = 0$. Both must be fixed before any large-scale reproduction is possible.

Observation 6 — RI cleaning is essential. With $\gamma = 1$, $K = 1000$, and rank-$r = 32$ SVD compression on Tiny-ImageNet, the accumulated regularisation bias is non-negligible: without RI cleaning, accuracy is 54.94%; applying RI cleaning improves this to 55.00%. The paper's ablation confirms that without RI, accuracy can drop by up to 9% for $\gamma = 100$.

Observation 7 — Saving scales linearly with K. The aggregate saving $\Delta B_{\text{total}} = K \cdot \Delta B_{\text{per-client}}$ scales perfectly linearly with the number of clients, making the compression increasingly valuable at larger scales.

X. Future Work Directions

1. Gram matrix compression via sketching. With $\mathbf{C}_k^r$ dominating 71.9% of the upload cost, the most impactful future direction is compressing $\mathbf{C}_k^r$. One approach is random sketching: transmit $\mathbf{S}\mathbf{C}_k^r\mathbf{S}^\top$ where $\mathbf{S} \in \mathbb{R}^{m\times d}$ is a Gaussian sketch matrix with $m \ll d$. A major open question is whether the AA law can be adapted to operate on sketched Gram matrices while preserving the data-partitioning invariance.

2. Adaptive rank selection per client. Rather than a fixed global r, each client could select the rank adaptively:

$$r_k^* = \min\left\{ r : \frac{\sum_{i=1}^r \sigma_i^2}{\sum_j \sigma_j^2} \geq \tau \right\}, \tag{27}$$

for a target energy threshold $\tau \in (0,1)$. This would yield heterogeneous compression ratios, with heavily non-IID clients achieving much higher compression.

3. Quantisation of SVD factors. Further communication reduction can be achieved by quantising the transmitted factors from float64 to lower precision (e.g., float16 or int8). The singular values $\mathbf{s}_r$ could be transmitted in float32 without significant loss.

4. Partial participation and straggler robustness. Extending the AA law to the partial participation setting where only a subset $\mathcal{S} \subset [K]$ of clients uploads their solution is a promising research direction.

5. Multi-layer analytic fine-tuning. Applying the analytic least-squares framework layer-by-layer using label projection techniques [14] would extend AFL to fine-tuning deeper portions of the backbone, potentially improving accuracy on domain-shifted tasks.

6. Differential privacy analysis of SVD factors. Transmitting SVD factors rather than the raw weight matrix may offer differential privacy advantages, since the low-rank factorisation discards directions that may encode individual training samples. A formal DP analysis would strengthen the privacy guarantees.

XI. Conclusion

This report presents a complete reproduction and novelty extension of the Analytic Federated Learning (AFL) framework proposed by He et al. [1].

We first identified and corrected three concrete implementation bugs in the released codebase: (i) a singular matrix inversion issue when $\gamma = 0$, for which we propose an automatic data-driven regularisation guard; (ii) a non-terminating infinite loop in the Dirichlet data partition routine, fixed by moving the minimum-size check outside the inner class loop; and (iii) a completely absent implementation of the claimed rank-r compression, which we implement from scratch using truncated SVD.

We reproduced AFL accuracy on CIFAR-100 (58.50% vs. paper's 58.56%) and Tiny-ImageNet (54.78% at $K = 100$ vs. paper's 54.67%) across client scales $K \in \{100, 500, 1000\}$, confirming the data-partitioning invariance property: performance remains constant regardless of the number of clients or the degree of data heterogeneity.

Our proposed novelty—rank-r truncated SVD compression of $\hat{\mathbf{W}}_k^r$ with $r = 32$—achieves a 77.7% reduction in weight matrix upload on Tiny-ImageNet (607 MB saved across 1000 clients) and a 61.7% reduction on CIFAR-100, while maintaining accuracy within 0.3% of the uncompressed baseline. A detailed communication cost analysis reveals that the Gram matrix $\mathbf{C}_k^r$ remains the dominant bottleneck (71.9% of per-client payload), motivating future work on $\mathbf{C}$ compression via sketching, quantisation, and adaptive rank selection.

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